Hasan Bukhari

Systems Programming • Embedded Systems • Low-Level Software

+1 601 913 4692 | hasan.bukhari25@gmail.com | LinkedIn | GitHub | Portfolio

Education

University of Southern Mississippi — **Expected Dec 2028**
B.S. in Computer and Information Sciences; Minor in Biological Sciences — *Hattiesburg, MS*

GPA: 4.00 • President's List • Academic Excellence Scholar

Beaconhouse College Program — **May 2024**
*Cambridge A Levels: 4 A*s in Biology, Chemistry, Mathematics, and Physics* — *Karachi, Pakistan*

Technical Experience

Data Analyst Intern, Health, Safety & Environment — **Jul 2026 – Aug 2026**
Engro Technical Services — *Karachi, Pakistan*

- Engineered Python/FastAPI automation, risk-scoring, reporting, and document-monitoring workflows across approximately 1,750 engineering/HSE files and 130,000+ operational records.
- Built production-facing analytical pipelines, Power BI models, DAX measures, and scheduled processes for operational decision support.

Student Worker, Operations Technology & Event Support — **Jun 2024 – Present**
Thad Cochran Center / Union Operations, University of Southern Mississippi — *Hattiesburg, MS*

- Build and support operational software for event requests, equipment tracking, inventory auditing, and university facility workflows.

Software Engineering Intern — **Aug 2023 – Feb 2024**
Shahid I.T. Consultants — *Karachi, Pakistan*

- Automated recurring record-update workflows in Python, reducing manual processing by 30%; supported integration testing, deployment documentation, and application maintenance.

Selected Systems & Embedded Projects

WCRE — Windows Checkpoint/Restore Engine | Rust, Win32 API, Windows PSS, x64 | GitHub — **2026**

- Developing an experimental user-space checkpoint/restart engine for Windows x64 that captures supported process state into persistent, integrity-checked checkpoint images and reconstructs controlled virtual-address layouts.
- Implemented process inspection, virtual-memory enumeration, PSS-backed snapshots, thread/register-subset capture, mapped-image inventory, checkpoint validation, restore planning, and exact reservation/commitment of selected PRIVATE memory ranges.

ESPCube — ESP32-S3 Handheld Interface | C++, ESP32-S3, PlatformIO, BLE, Wi-Fi/TCP, Rust | GitHub — **2026**

- Built embedded firmware integrating a touchscreen, 6-axis IMU, physical controls, microphones, audio hardware, Bluetooth HID, custom BLE services, Wi-Fi, and PSRAM-backed audio buffering.
- Designed device/host communication using HID, BLE/GATT, ADPCM speech transport, and TCP PCM audio streaming; built a native Rust Windows companion for BLE lifecycle, local speech inference, system-audio capture, and native input.

AEGIS — Local-First Windows Document Platform | Python, FastAPI, Rust/Tauri, React | GitHub — **2026**

- Engineered filesystem monitoring, deterministic version diffing, SHA-256 integrity checks, append-only audit history, adversarial tests, and native Windows packaging.

To Ash Again — 2D Narrative Platformer | Python, Pygame, OOP | GitHub — **2025**

- Developed a four-act standalone game over two months with modular gameplay systems, enemy AI, character switching, inventory/powers, save state, cutscenes, and multi-phase boss logic.

Technical Skills

Languages: Rust, C++, Python, PowerShell, SQL; Kotlin, JavaScript/TypeScript
Systems: Win32 API, Windows PSS, virtual memory, process/thread inspection, x64 register state, filesystem monitoring, TCP/IP
Embedded: ESP32-S3, PlatformIO, Bluetooth HID, BLE/GATT, Wi-Fi, I2C, GPIO, PSRAM, PCM/audio streaming
Tools: Cargo, Git/GitHub, GitHub Actions, Linux/WSL, Windows development, FastAPI, Tauri, native packaging and release tooling

Leadership & Honors

President & Founder, Student Poets Association and Creative Writing Club, USM — Lead arts programming, public showcases, open mics, and organizational operations.

Selected Honors: Outstanding Cambridge Learner Award; ITC 371 Best Project & Best Assistant Project Manager; UGS Best Art as Research Project (2026); Best Arts Showcase for Poetry (2025, 2026); President's List.